INCH-POUND
MIL-DTL-24652A(SH)
1 November 2016
SUPERSEDING
MIL-E-24652(SH)
28 May 1985

DETAIL SPECIFICATION

EXTINGUISHER, FIRE, AQUEOUS FILM FORMING FOAM (AFFF), 2.5-GALLON

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a 2.5-gallon fire extinguisher of the stored-pressure type. It is capable of generating a low expansion foam using Aqueous Film Forming Foam (AFFF) concentrate and potable water.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-H28	-	Screw-Thread Standards for Federal Services
FED-STD-H28/2	-	Screw-Thread Standards for Federal Services, Section 2, Unified Inch Screw Threads - UN and UNR Thread Forms

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-S-901	-	Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for
MIL-F-24385	-	Fire Extinguishing Agent, Aqueous Film Forming Foam (AFFF) Liquid Concentrate, for Fresh and Sea Water

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard, DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II – Internally Excited)

(Copies of these documents are available online at http://quicksearch.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at http://www.asq.org.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 11 - Standard for Low-, Medium-, and High-Expansion Foam

(Copies of this document are available online at www.nfpa.org.)

SAE INTERNATIONAL

SAE AMS-STD-595 - Colors Used in Government Procurement

(Copies of this document are available online at http://www.sae.org.)

UNDERWRITERS LABORATORIES, INC. (UL)

UL 8 - Water Based Agent Fire Extinguishers

UL 162 - Foam Equipment and Liquid Concentrates

UL 711 - Rating and Fire Testing of Fire Extinguishers

(Copies of these documents are available online at http://ul.com.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials.

3.2.1 Shell (cylinder). The shell shall be fabricated of austenitic stainless steel, American Iron and Steel Institute (AISI) Type 304L.

3.2.1.1 Finish. The shell shall be finished with a red enamel protective coating, such as polyester or polyurethane, for identification and enhanced corrosion resistance. The coating shall be color number 11105 OSHA Safety Red/DOT Red in accordance with SAE AMS-STD-595.

3.2.2 Discharge valve. The valve body and operating parts (levers, handle, valve stem, plunger pin, springs, bushings, etc.) shall be made of nonferrous metal or austenitic stainless steel.

3.2.3 Mounting bracket. The mounting bracket shall be of metallic construction sufficient for its intended use. The mounting bracket shall be finished with an enamel protective coating, such as polyester or polyurethane, for enhanced corrosion resistance.

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3.2.4 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Fire extinguishing agent. The fire extinguishing agent tested for listing by Underwriters Laboratories Inc. (UL) shall be a solution of liquid AFFF concentrate and fresh water. The fire extinguisher shall be compatible with the use of AFFF in accordance with MIL-F-24385.

3.4 Weight. The weight of the empty extinguisher, excluding the mounting bracket, shall not exceed 10 pounds.

3.5 Overall dimensions. The extinguisher and bracket shall be as small as possible consistent with good design and performance. The extinguisher height shall be not greater than 25 inches.

3.6 Design and construction.

3.6.1 Extinguisher. The extinguisher shall be a stored-pressure type with a rated capacity of 2.5 gallons of AFFF solution and a service pressure of 100 pounds per square inch (lb/in^2). The extinguisher shall be easily rechargeable without the use of special tools or special equipment. (Special tools are defined as those not listed in the Federal Supply Catalog, copies of which may be consulted in the office of the Defense Contract Administration Service.) The flow shall be controlled by a shut off valve at the extinguisher end of the discharge hose. The extinguisher shall be listed by UL in accordance with UL 8 and shall have a minimum fire extinguishing rating of 3-A:20-B.

3.6.2 Extinguisher components. The extinguisher shall contain the following major components:

a. Discharge valve assembly (see 3.6.2.1).

b. Discharge hose and nozzle assembly (see 3.6.2.2).

c. Shell (see 3.6.2.3).

d. Siphon tube and strainer assembly (see 3.6.2.4).

e. Anti-overfill tube (see 3.6.2.5).

3.6.2.1 Discharge valve assembly. The discharge valve assembly shall consist of a discharge valve, air valve assembly, pressure gauge, and pull pin and retainer chain.

3.6.2.1.1 Discharge valve. A self-closing discharge valve shall be located on the top of the extinguisher.

3.6.2.1.1.1 Gaskets and O-rings. A gasket or O-ring shall be retained in a recess, or the equivalent, in the valve or collar to provide a leak-tight seal. A gasket or O-ring shall be provided at the valve and hose connection to provide a leak-tight joint. Gaskets or O-rings shall be used to prevent leaking of the discharge valve at the seat and along the shaft.

3.6.2.1.1.2 Threads. Threads shall be in accordance with FED-STD-H28 and FED-STD-H28/2. The discharge valve connection to the shell shall have 1.5-16 UN-2B (internal) threads. The discharge valve connection to the discharge hose shall have 0.625-18 UNF-2B (internal) threads.

3.6.2.1.2 Air valve assembly. A Schraeder type (or equal) air valve assembly with a stainless steel or nonferrous metallic cap shall be provided as an integral part of the discharge valve assembly for pressurizing the extinguisher. The cap shall be held captive to the extinguisher by a short stainless steel or nonferrous metallic retainer chain, so that the cap will not be lost upon removal. The cap retainer chain shall be attached to a stationary part of the valve via a stainless steel or nonferrous metallic fastener dedicated solely to this purpose. Pressure charging through the discharge hose is not acceptable.

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3.6.2.1.3 Pull-pin and retainer chain. A pull-pin shall be provided to prevent inadvertent operation. The pin shall pass through the valve operating lever in such a way that the valve will not operate with the pin in place. The pin shall be held captive to the extinguisher by a short stainless steel or nonferrous metallic retainer chain, so that the pin will not be lost during operation of the unit. The pull-pin retainer chain shall be attached to the stationary part of the valve handle via a stainless steel or nonferrous metallic fastener dedicated solely to this purpose. A visual mechanical tamper seal shall be used to hold the pull-pin in the safe position. Use of or tampering with the extinguisher will break the seal. The seal is to be field-replaceable and shall be provided as a part of the charging kit (see 3.6.3).

3.6.2.1.4 Gauge. A pressure gauge shall be provided as an integral part of the discharge valve. The gauge face shall indicate the pressure in units of lb/in² or a combination of lb/in² and SI units, such as kilopascals (kPa). The minimum indicated gauge pressure shall be 0 lb/in². The maximum indicated gauge pressure shall be not less than 150 lb/in² and not greater than 250 lb/in². The gauge shall be color-coded to indicate operable pressure range. The diameter of the gauge shall be not less than 1.5 inches.

3.6.2.2 Discharge hose and nozzle assembly. The extinguisher shall have a flexible discharge hose with a permanently affixed nozzle. The inlet side of the discharge hose shall contain a fitting with 0.625-18 UNF-2A (external) threads for attachment to the discharge valve. The nozzle shall be of the air-aspirating type and shall aerate the extinguishing agent to achieve the expansion ratio specified in 3.7.1.

3.6.2.3 Shell. The shell (body) of the extinguisher shall be of sufficient volume to hold 2.5 gallons of the extinguishing agent plus enough compressed air to propel the agent at velocities sufficient to aerate the agent and produce the discharge characteristics specified in 3.7.1. The shell outside diameter shall be 7±0.125 inches. The design service pressure shall be not less than 100 lb/in².

3.6.2.3.1 Joints. Joints shall be butt welded. Repair welds are not acceptable.

3.6.2.3.2 Fill opening. The collar or fill opening of the shell shall have 1.5-16 UN-2A (external) threads.

3.6.2.4 Siphon tube and strainer assembly. A siphon tube with strainer shall be attached to the discharge valve and shall carry agent from the bottom of the shell to the discharge valve.

3.6.2.5 Anti-overfill tube. An anti-overfill tube shall be provided. It shall be capable of ensuring adequate air space above the liquid in the extinguisher so that the propellant air will expel the liquid at sufficient pressure to empty the extinguisher and discharge foam with the desired characteristics (see 3.7.1) throughout the duration of extinguisher operation.

3.6.3 Charging kit. Extinguishers shall be shipped uncharged and shall contain the following items packed as a charging kit:

a. Complete instructions for charging with 1 percent, 3 percent, and 6 percent foam concentrates. The instructions shall require mixing the concentrate with potable water having less than 40 parts per million chlorides.

b. Empty premeasured container for measuring a charge of AFFF concentrate. Container demarcation shall be in ounces and shall be sufficient for use with 1 percent, 3 percent, and 6 percent foam concentrates.

c. Plastic tamper seal.

d. Extinguisher inspection record tag with suitable attachment.

3.6.4 Mounting bracket. A mounting bracket shall be provided to hold the extinguisher in a vertical orientation and shall be capable of wall mounting. The mounting bracket shall have a bottom plate which extends beyond the base of the extinguisher.

3.7 Performance requirements. The fire extinguisher shall exhibit the performance requirements specified herein during the testing outlined in 4.5.

3.7.1 Discharge characteristics. The effective discharge time of the extinguisher shall be 45 to 65 seconds when tested as specified in 4.5.2.1. The foam shall have a minimum expansion ratio of 6.5:1 and a minimum 25 percent drainage time of 90 seconds when tested as specified in 4.5.2.2.

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3.7.2 Fire extinguisher.

3.7.2.1 UL rating. The extinguisher shall have a fire extinguishing capability so as to obtain a 3-A:20-B fire extinguishing rating from UL and shall be so labeled.

3.7.2.2 Fire extinguishing. The extinguisher shall extinguish the fire when tested as specified in 4.5.4.

3.7.2.2.1 Equivalency. The requirement of 3.7.2.2 shall be considered to have been met if the foam tested by UL to obtain the fire extinguishing rating (see 3.7.2.1) is qualified under MIL-F-24385.

3.7.3 Repeated operation of the valve. When tested as specified in 4.5.3.1, repeated operation of the valve shall not cause undue wear or adversely affect the operation of the extinguisher.

3.7.4 Repeated operation of the extinguisher mounting bracket. The mounting bracket shall be operable after the test specified in 4.5.3.2.

3.7.5 High-impact and shock resistance. The extinguisher and mounting bracket shall be operable after being subjected to the high-impact shock tests specified in 4.5.5.1.

3.7.6 Vibration resistance. The extinguisher and mounting bracket shall be operable after being subjected to the vibrational conditions specified in 4.5.5.2.

3.8 Identification.

3.8.1 Nameplate. A nameplate and other extinguisher marking shall be as specified in UL 8.

3.8.2 Military label plate. A label plate shall be securely attached to the body of each extinguisher on the rear side (opposite to the nameplate). The label shall be an aluminum foil type with pressure sensitive adhesive backing. The label plate shall be legibly printed with black lettering at least 0.25 inch high and shall contain the following information:

a. “MIL-DTL-24652”.

b. National Stock Number (NSN), if provided (see 6.2).

c. Allowance Parts List (APL), number if provided (see 6.2).

d. “FOLLOW APPLICABLE PMS INSTRUCTIONS ON INSPECTION, MAINTENANCE, AND RECHARGE.”

3.9 Workmanship. The extinguishers shall be uniform in quality and shall be free from irregularities, defects, or foreign matter which could affect safety, performance, reliability, or durability.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

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4.2 First article inspection. First article inspection shall be performed on the sample extinguishers (see 4.3.2) when a first article sample is required (see 3.1). This inspection shall include the examination of 4.4 and the tests specified in table I.

TABLE I. First article inspection.

Test	Requirement paragraph	Test paragraph
Visual examination	3.2 – 3.2.3, 3.7.2.1, 3.8, 3.9	4.4
Discharge duration	3.7.1	4.5.2.1
Foam quality	3.7.1	4.5.2.2
Fire extinguishing	3.7.2.2	4.5.4
Discharge valve operation	3.7.3	4.5.3.1
Mounting bracket operation	3.7.4	4.5.3.2
Shock	3.7.5	4.5.5.1
Vibration	3.7.6	4.5.5.2

4.3 Conformance inspection. Conformance inspection shall include the examination of 4.4.

4.3.1 Inspection lot. All extinguishers offered for delivery at one time shall be considered a lot for the purpose of conformance inspection.

4.3.2 Sampling for examination. Sample extinguishers shall be selected at random from each lot in accordance with ANSI/ASQ Z1.4, at inspection level S-1.

4.4 Examination. This element of inspection shall encompass all visual examinations and dimensional measurements. Each of the sample extinguishers selected in accordance with 4.3.2 shall be examined to verify conformance to this specification. Examination shall be conducted to determine defects as specified in table II. Noncompliance with any specified requirements or presence of one or more defects shall constitute cause for rejection.

TABLE II. Classification of defects.

Categories	Defects	Requirements paragraphs
Critical: None determined.		
Major:		
101	Material not as specified.	3.2
102	Component missing, damaged, defective foreign matter present, irregularities, or malfunctioning.	3.6.1 – 3.2.3, 3.9
103	Assembly of parts not as specified.	3.6.1 – 3.2.3, 3.9
104	Threading not as specified.	3.6.2.1.1.2, 3.6.2.2, 3.6.2.3.2
105	Finish of exterior surfaces not as specified.	3.2.1.1, 3.2.3, 3.9
106	Dimensions not as specified.	3.5, 3.6.2.1.4, 3.6.2.3
107	UL rating not as specified.	3.7.2.1
Minor:		
201	Weight not as specified.	3.4
202	Marking for identification not as specified.	3.8

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4.5 Test procedures.

4.5.1 General. The tests of 4.5.2.1, 4.5.2.2, and 4.5.4 shall be conducted with the extinguisher charged in accordance with the manufacturer's instructions with Type 3 or Type 6 AFFF concentrate in accordance with MIL-F-24385.

4.5.2 Discharge. Discharge and foam quality tests shall be conducted in accordance with NFPA 11 and as specified in 4.5.2.1 and 4.5.2.2. The tests shall be conducted at 70±5 °F.

4.5.2.1 Discharge duration. An extinguisher charged to its rated capacity (see 4.5.1) and rated pressure (see 3.6.1) and conditioned at 70±5 °F for 16 hours shall be held upright with the cylinder in a vertical position. The discharge nozzle shall be held in a stationary horizontal position at a height of 3 feet above the floor. The extinguisher shall be discharged in a continuous stream for a minimum of 45 seconds. Maximum reach at the beginning and end of this time span shall be recorded. The reach shall be defined as the distance from the nozzle to the point on the floor at which 85 percent of the discharged mass lies beyond the point. The extinguisher shall continue to be discharged until the "gas point" is reached. The total duration of this discharge shall be recorded. The "gas point" occurs when the discharged foam abruptly changes quality, indicating that air is being drawn from the tank along with the liquid. Failure to meet the discharge time criteria of 3.7.1 shall constitute failure of this test.

4.5.2.2 Foam quality. The foam quality test specified in NFPA 11 shall be conducted twice (see 4.5.1) for each extinguisher, once with a full charge and once after expelling half the liquid from the unit. The averaged results of the two tests shall be recorded as expansion ratio and 25 percent drainage time (in seconds) as shown in NFPA 11. Failure to meet the expansion ratio and 25 percent drainage time criteria of 3.7.1 shall constitute failure of this test.

4.5.3 Operational tests.

4.5.3.1 Discharge valve operation. The valve shall be operated 100 times with the valve pressurized to 100±20 lb/in^2. Operation shall be normal at the completion of the test.

4.5.3.2 Mounting bracket operation. The extinguisher shall be locked in the mounting bracket, and the bracket shall be opened and relocked. This procedure shall be repeated 100 times. Operation of the bracket shall be normal at the completion of this test.

4.5.4 Fire extinguishing.

4.5.4.1 General. The test fire and test protocol shall be in accordance with the UL 711 flammable liquid fire test for Class B extinguishers. Failure to extinguish the test fire required for a 20-B rating shall constitute failure of this test.

4.5.4.2 AFFF agent. The extinguisher shall be filled to its rated capacity with 3±0.05 percent of Type 3 or 6±0.1 percent of Type 6 AFFF/potable water solutions using AFFF qualified under MIL-F-24385.

4.5.5 Environmental tests.

4.5.5.1 Shock. The extinguisher and mounting bracket shall be subjected to shock testing for Grade A, Class I, Type A lightweight equipment, for both ships and submarines, as specified in MIL-S-901. The configuration of the test shall have the mounting bracket attached to the test fixture in the wall mount configuration with a full charged extinguisher locked in the mounting bracket. At the completion of the shock test, the bracket shall have retained the extinguisher without damaging the extinguisher, the bracket shall be normally operable, the extinguisher shall be normally operable with no damage that may limit operation, and the extinguisher shall not have leaked. At the end of the shock test, the extinguisher shall stand for a minimum of 10 minutes. It shall then be held in a vertical orientation and discharged completely into a container, as evidenced by cessation of flow from the nozzle. Not less than 80 percent (by weight) of the rated agent capacity shall have been discharged.

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4.5.5.2 Vibration. The extinguisher and mounting bracket shall be subjected to the Type I vibration test, for both ships and submarines, specified in MIL-STD-167-1 for shipboard equipment. The configuration of the test shall have the mounting bracket attached to the test fixture in the wall mount configuration with a full charged extinguisher locked in the mounting bracket. At the completion of the vibration test, the bracket shall have retained the extinguisher without damaging the extinguisher, the bracket shall be normally operable, the extinguisher shall be normally operable with no damage that may limit operation, and the extinguisher shall not have leaked. At the end of the vibration test, the extinguisher shall be tested for percent discharge as specified in 4.5.5.1. Not less than 80 percent (by weight) of the rated agent capacity shall have been discharged.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The fire extinguishers covered by this specification are intended for use on naval surface ships and submarines on Class A (common solid combustible) and Class B (flammable liquid) fires. They are not intended for use on Class C (electrical) or Class D (combustible metal) fires. Existing stock of fire extinguishers manufactured to MIL-E-24652 dated 28 May 1985 are acceptable for use until depleted.

6.1.1 Extinguishing media. The extinguishers are intended to be used with a solution of AFFF appropriate for the ship class (conforming to Type 3 or Type 6 of MIL-F-24385, or 1-percent UL 162) and potable water.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. When first article is required and applicable procedures (see 3.1 and 6.3).

c. NSN and APL number (see 3.8.2).

d. Packaging requirements (see 5.1).

6.3 First article. When a first article inspection is required (see 6.2), the first article sample should consist of one unit. Specific instructions should be included in the acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 Subject term (key word) listing.

Firefighting equipment

Portable

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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